

Collections – List and Sets

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1

```
class SwapValues {
```

```
    public static <T> void swap(T a, T b) {
```

```
        System.out.println("First = " + a);
```

```
        System.out.println("Second = " + b);
```

```
        T temp = a;
```

```
        a = b;
```

```
        b = temp;
```

```
        System.out.println("After Swapping:");
```

```
        System.out.println("First = " + a);
```

```
        System.out.println("Second = " + b);
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        System.out.println("Integer Values");
```

```
        swap(10, 20);
```

```
System.out.println();
```

```
System.out.println("String Values");
```

```
swap("hello", "world");
```

```
}
```

```
}
```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and a user profile for SELVA PRABHU A K. The main section is titled 'Class Activity - Generic methods and Class'. On the left, a sidebar lists three questions: Q1 (Swap Two Values, 10 marks), Q2 (Find the Maximum Mark, 10 marks), and Q3 (Generic Class - Product Inventory Holder, 10 marks). The main area displays Q1, which asks to write a Java program that creates a generic method named swap() to interchange two values. The question specifies creating a generic method swap(T a, T b), calling it with integer and string values, and displaying the values before and after swapping. Below the question, there is a 'Your Code' section with a code editor containing the following Java code:

```
1 class SwapValues {
2
3     public static <T> void swap(T a, T b) {
4         System.out.println("First = " + a);
5         System.out.println("Second = " + b);
6
7         T temp = a;
8         a = b;
9         b = temp;
10
11         System.out.println("After Swapping:");
12         System.out.println("First =" + a);
```

To the right of the code editor is an 'Input' field with 'stdin input' and an 'Output' window showing the following text:

```
Integer Values
First = 10
Second = 20
After Swapping:
First =20
Second=10
```

2

```
class demo {
```

```
    public static <T extends Comparable<T>> void compa(T a, T b, T c) {
```

```
        if (a.compareTo(b) > 0 && a.compareTo(c) > 0) {
```

```
            System.out.println(a);
```

```
        }
```

```
        else if (b.compareTo(a) > 0 && b.compareTo(c) > 0) {
```

```
            System.out.println(b);
```

```
        }
```

```
        else {
```

```
            System.out.println(c);
```

```
        }
```

```
    }
```

```

public static void main(String args[]) {

    System.out.println("Integer");

    compa(60, 70, 80);

    System.out.println("Double");

    compa(60.8, 70.9, 40.8);

}

}

```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user "SELVA PRABHU A K" with ID "192421224RA". The page is for "JAVA PROGRAMMING".

On the left, a "QUESTIONS" sidebar lists three questions:

- Q1: Swap Two Values (10 marks)
- Q2: Find the Maximum Mark (10 marks)
- Q3: Generic Class - Product Inventory Holder (10 marks)

 It indicates "3/3 answered".

The main area displays "Q2 Find the Maximum Mark" (10 marks). The question text is:

A college wants a reusable method to find the highest mark among different numeric types (Integer, Double, Float). Write a Java program that creates a generic method findMax() to return the largest of three values.

<T> T findMax(T a, T b, T c)

Accept three values of the same type.

Display the maximum value.

Test the method using:

Integer marks

Double marks

Below the question, there is a "Your Code" section with a code editor showing a Java program:

```

1 class demo{
2     public static <T>void compa(Ta,Tb,Tc) {
3         if(a>b&&a>c){
4             System.out.println(a);
5         }
6         else if(b>a&&b>c){
7             System.out.println(b);
8         }
9         else{
10            System.out.println(c);
11        }
12    }
13 }

```

To the right of the code editor is an "Input" field with "stdin input" and an "Output" section showing a Java compiler error:

```

java: error: <identifier>
expected
    public static <T>void
    compa(Ta,Tb,Tc){
    ~
    }

```

3

```

class Inventory<T> {

    T item;

    Inventory(T item) {

        this.item = item;

    }

}

```

```
void showItem() {  
    System.out.println(item);  
}  
  
public static void main(String[] args) {  
  
    Inventory<Integer> id = new Inventory<>(101);  
    Inventory<String> name = new Inventory<>("Wireless Mouse");  
    Inventory<Double> price = new Inventory<>(799.50);  
  
    System.out.print("Product ID: ");  
    id.showItem();  
  
    System.out.print("Product Name: ");  
    name.showItem();  
  
    System.out.print("Product Price: ");  
    price.showItem();  
}  
}
```

Product Price: 799.50

Your Code

```
1 class Inventory<T> {  
2     T item;  
3     Inventory(T item) {  
4         this.item = item;  
5     }  
6     void showItem() {  
7         System.out.println(item);  
8     }  
9     public static void main(String[] args) {  
10  
11         Inventory<Integer> id = new Inventory<> (101);  
12         Inventory<String> name = new Inventory<> ("Wireless Mouse");  
13         Inventory<Double> price = new Inventory<> (799.50);  
14  
15         System.out.print("Product ID: ");  
16         id.showItem();  
17  
18         System.out.print("Product Name: ");  
19         name.showItem();  
20  
21         System.out.print("Product Price: ");  
22         price.showItem();  
23     }
```

Input

stdin input

Output

```
Product ID: 101  
Product Name: Wireless Mouse  
Product Price: 799.5
```